

Unit A: Nervous and Endocrine Systems

Themes: Equilibrium and Systems

Overview: This unit examines the biological processes that mediate the interactions between humans and their environment to maintain equilibrium. The nervous system contributes to homeostasis through its response to internal and external stimuli. Endocrine glands help to maintain homeostasis through the hormones they release into the blood. A study of the interactions between the nervous and endocrine systems leads to an examination of the functioning of the central and peripheral nervous systems and their ability to sense the environment and respond to it.

This unit builds on:

- Grade 8 Science, Unit B: Cells and Systems
- Science 10, Unit C: Cycling of Matter in Living Systems
- Biology 20, Unit D: Human Systems

Unit A will require approximately 25% of the time allotted for Biology 30.

Links to Mathematics: Refer to page 56.

Focusing Questions: How does the human body maintain equilibrium between its internal and external environments? What physiological processes and control systems are involved in maintaining homeostasis? What medical technologies are available to alleviate disorders of the nervous and endocrine systems?

General Outcomes: There are two major outcomes in this unit.

Students will:

1. explain how the nervous system controls physiological processes
2. explain how the endocrine system contributes to homeostasis.

Key Concepts: The following concepts are developed in this unit and may also be addressed in other units or in other courses. The intended level and scope of treatment is defined by the outcomes.

- neuron
- nerve impulse transmission
- central and peripheral nervous systems
- reflex arcs
- sensory receptors
- endocrine system and hormones
- homeostasis and feedback systems
- endocrine and nervous system interactions

General Outcome 1

Students will explain how the nervous system controls physiological processes.

Specific Outcomes for Knowledge

- Students will:*
- 30–A1.1k describe the general structure and function of a neuron and myelin sheath, explaining the formation and transmission of an action potential, including all-or-none response and intensity of response; the transmission of a signal across a synapse; and the main chemicals and transmitters involved, i.e., norepinephrine, acetylcholine and cholinesterase
 - 30–A1.2k identify the principal structures of the central and peripheral nervous systems and explain their functions in regulating the voluntary (somatic) and involuntary (autonomic) systems of the human organism; i.e., cerebral hemispheres and lobes, cerebellum, pons, medulla oblongata, hypothalamus, spinal cord, sympathetic and parasympathetic nervous systems, and the sensory-somatic nervous system
 - 30–A1.3k describe, using an example, the organization of neurons into nerves and the composition and function of reflex arcs; *e.g., the patellar reflex, the pupillary reflex*
 - 30–A1.4k describe the structure and function of the parts of the human eye; i.e., the cornea, lens, sclera, choroid, retina, rods and cones, fovea centralis, pupil, iris and optic nerve
 - 30–A1.5k describe the structure and function of the parts of the human ear, including the pinna, auditory canal, tympanum, ossicles, cochlea, organ of Corti, auditory nerve, semicircular canals and Eustachian tube
 - 30–A1.6k explain other ways that humans sense their environment and their spatial orientation in it; *e.g., olfactory receptors, proprioceptors, taste receptors, receptors in the skin.*

Specific Outcomes for Science, Technology and Society (STS) (Nature of Science Emphasis)

- Students will:*
- 30–A1.1sts explain that scientific knowledge and theories develop through hypotheses, the collection of evidence, investigation and the ability to provide explanations (**NS2**)
 - *discuss the biological basis of neurological diseases such as Alzheimer's or Parkinson's disease and how this relates to treatment*
 - *evaluate the impact of photoperiod (light wavelength and duration) on humans and identify adaptations to light deprivation in northern communities*
 - 30–A1.2sts explain that scientific investigation includes the process of analyzing evidence and providing explanations based upon scientific theories and concepts (**NS5f**) [**ICT C7–4.2**]
 - *analyze experimental evidence regarding the influence of anesthetics, drugs and chemicals, natural and synthetic, on the functioning of the nervous system and relate this to addiction theories*
 - *analyze the contribution of technological developments and physiological knowledge to longevity and quality of life*
 - 30–A1.3sts explain that the goal of technology is to provide solutions to practical problems (**ST1**) [**ICT F2–4.4**]
 - *investigate technologies available to correct eye defects such as myopia, hyperopia and astigmatism and ear defects such as hearing loss and tinnitus*
 - *investigate the biological basis of neurotoxin action and their antidotes (snake venom, box jellyfish, botulin, reserpine [*Rauwolfia serpentina*])*
 - *investigate the use of neurotoxins by Indigenous peoples*
 - *discuss how advances in science have contributed to technologies that increase access to the world beyond normal sensory limits.*

Note: Some of the outcomes are supported by examples. The examples are written in italics and **do not form part of the required program** but are provided as an illustration of how the outcomes might be developed.

General Outcome 1

Students will explain how the nervous system controls physiological processes.

Specific Outcomes for Skills (Nature of Science Emphasis)

Initiating and Planning

- 30–A1.1s *Students will:*
formulate questions about observed relationships and plan investigations of questions, ideas, problems and issues
- design an experiment to investigate heat, cold, pressure and touch receptors (**IP–NS1, IP–NS2, IP–NS3**).

Performing and Recording

- 30–A1.2s *Students will:*
conduct investigations into relationships between and among observable variables and use a broad range of tools and techniques to gather and record data and information
- design and perform an experiment to investigate the physiology of reflex arcs (**IP–NS2, PR–NS2, PR–NS3, PR–NS4**) [**ICT F1–4.2**]
 - perform experiments to measure the ability to discriminate objects visually and to hear a range of sounds (**PR–NS2, PR–NS3, PR–NS4**)
 - use a microscope and prepared slides to observe neurons and synapses (**PR–NS2, PR–NS3, PR–NS4**)
 - observe the principal features of a mammalian brain, eye and ear, using models, computer simulations or dissections, and identify the major structures of those organs (**PR–NS3, PR–NS4**) [**ICT C6–4.4**]
 - *investigate and integrate, from library and electronic sources, information on the impact of photoperiod and wavelength on humans* (**PR–NS1, PR–NS4**) [**ICT C1–4.1**]
 - *compile and display, in appropriate format, data collected from investigations on reflex arcs, stimulus strength versus force of muscle contraction, and/or auditory range* (**PR–NS4**) [**ICT P2–4.1**].

Analyzing and Interpreting

- 30–A1.3s *Students will:*
analyze data and apply mathematical and conceptual models to develop and assess possible solutions
- *interpret patterns and trends in data on strength of stimuli versus force of muscle contraction* (**AI–NS2**) [**ICT C7–4.2**]
 - *analyze and interpret aspects of vision, such as blind spot, acuity, accommodation, adaptation, binocular vision and peripheral vision* (**AI–NS2**) [**ICT C7–4.2**]
 - *analyze a hearing aid as a device that simulates a sensory function* (**AI–ST1**)
 - *pose new questions, such as: “Why are some people more tolerant to pain than others?”* (**AI–NS5**)
 - *collect and analyze class data on colour charts* (**PR–NS4, AI–NS2**) [**ICT C7–4.2**]
 - *analyze data that shows the interrelationship between taste and smell receptors* (**AI–NS2**) [**ICT C7–4.2**].

Communication and Teamwork

- 30–A1.4s *Students will:*
work collaboratively in addressing problems and apply the skills and conventions of science in communicating information and ideas and in assessing results
- use appropriate Système international (SI) units, fundamental and derived units and significant digits (**CT–NS2**)–
 - use appropriate numeric, symbolic, graphical and linguistic modes of representation to communicate ideas, plans and results (**CT–NS2**)–
 - *work cooperatively with group members to investigate neurological disorders such as Alzheimer’s or Parkinson’s disease* (**CT–NS1**).
- To be developed throughout the course.

Note: Some of the outcomes are supported by examples. The examples are written in italics and **do not form part of the required program** but are provided as an illustration of how the outcomes might be developed.